

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

A Data-Driven Logistics Analytics and Interactive Dashboard Approach

Research Paper / Internship Project Paper

Prepared from the supplied Nassau Candy Distributor project files

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Abstract

Efficient factory-to-customer transportation is essential for controlling logistics cost, maintaining service quality, and supporting scalable distribution operations. This paper presents a data-driven analysis of Nassau Candy Distributor's shipping records to evaluate route efficiency, geographic performance, shipping-mode differences, and potential bottlenecks. The supplied dataset contains 10,194 shipment records, 8,549 unique orders, 15 products, and 59 customer state/province values. The methodology includes data validation, date parsing, missing-value assessment, product-to-factory mapping, lead-time calculation, factory-to-customer route construction, KPI development, exploratory data analysis, route ranking, and interactive visualization. A Streamlit dashboard was developed using Python, Pandas, NumPy, and Plotly to support route-level drill-down and operational monitoring. The analysis successfully maps all supplied products to five factories and produces 196 factory-to-state route combinations. A major data-quality issue was identified: order dates span 2024–2025 while ship dates span 2026–2030, producing calculated lead times of 904–1,642 days. Therefore, the results are appropriate for demonstrating an analytics workflow, but the date fields should be validated with the source system before operational decisions are made. The project provides a foundation for future predictive delivery modeling and route optimization.

1. Introduction

Modern distribution networks generate large volumes of transactional and shipment data. However, raw shipment records do not directly reveal which factory-to-customer routes perform well, where delays are concentrated, or which operational segments should receive management attention. A route-level analytical view can convert these records into actionable logistics intelligence.

This study focuses on Nassau Candy Distributor and examines factory-to-customer shipping performance using a supplied shipment dataset. The central analytical route definition is Factory → Customer State. Region-level analysis is used to provide a broader geographic perspective, while shipping-mode analysis compares Standard Class, Second Class, First Class, and Same Day services.

The study also emphasizes responsible data science. Instead of suppressing unusual values, the workflow identifies and documents a major date inconsistency in the source data. This validation step is important because misleading dates can produce apparently precise but operationally invalid conclusions.

2. Problem Statement

The distributor requires a systematic method to identify efficient and inefficient factory-to-customer routes, compare shipping performance across regions and shipping modes, and detect high-volume areas that may represent operational bottlenecks. Manual inspection of individual orders is not sufficient for this purpose.

The research problem is therefore defined as follows: How can shipment transaction data be transformed into route-level performance measures and an interactive decision-support dashboard while maintaining appropriate data-quality controls?

3. Objectives

The objectives of the project are:

1. Validate and clean order and shipment records.

2. Calculate shipment lead time from order and ship dates.
3. Map products to their supplying factories.
4. Construct and aggregate Factory → Customer State routes.
5. Measure average lead time, variability, volume, delay frequency, and route efficiency.
6. Compare geographic regions and shipping modes.
7. Provide interactive filtering and route drill-down.
8. Identify data-quality limitations and propose improvements.
9. Establish a foundation for predictive delivery analysis and route optimization.

4. Dataset and Data Preparation

The supplied CSV contains 10,194 records and 18 original fields. The dataset includes order identifiers, dates, ship mode, customer information, geography, product information, sales, units, gross profit, and cost. There are 8,549 unique orders, 15 products, and 59 customer state/province values.

Data preparation included parsing Order Date and Ship Date, checking missing values, calculating lead time, validating negative durations, mapping products to factories, standardizing route construction, and generating route- and mode-level summaries. No missing values were found in the original fields, no negative lead times were observed, and all 15 products were successfully mapped to a factory.

The project uses a supplied product-to-factory relationship. The five factories represented in the analysis are Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory.

5. Methodology

The analytical workflow follows a standard data-science pipeline.

5.1 Shipping Lead Time

Shipping Lead Time = Ship Date – Order Date.

5.2 Route Construction

Route = Factory → Customer State.

5.3 Delay Indicator

A shipment is treated as delayed when its lead time exceeds a user-selected threshold. The dashboard default is 1,300 days because the supplied dataset contains unusually large lead times; the threshold can be changed interactively.

5.4 Route Efficiency Score

A normalized 0–100 score is calculated from route average lead time:

Efficiency Score = $100 \times (\text{Maximum Route Average} - \text{Route Average}) / (\text{Maximum Route Average} - \text{Minimum Route Average})$.

Higher scores indicate lower observed average lead time.

5.5 Route Ranking

The dashboard supports a minimum shipment-count filter. This reduces the risk that a route with only one observation dominates a ranking simply because of sampling variation.

5.6 Geographic Analysis

Average lead time, shipment volume, and sales are summarized by customer state and region. Non-US locations are handled separately where the map geometry does not represent them.

5.7 Shipping-Mode Analysis

Standard Class, Second Class, First Class, and Same Day are compared using shipment volume, average lead time, median lead time, variability, delay frequency, and sales.

6. Results and Exploratory Analysis

The supplied dataset contains 10,194 shipment records. Standard Class represents the largest share of shipments, followed by Second Class, First Class, and Same Day.

Regional analysis shows that Gulf has the lowest observed average lead time among the four named regions, while Interior has the highest. Shipping-mode averages are relatively close in the supplied data, although their order mix and geographic composition may differ. These descriptive differences should therefore not be interpreted as causal evidence that one shipping mode is inherently better or worse.

The route summary contains 196 factory-to-state route combinations. Raw fastest and slowest routes can be dominated by very small shipment counts; consequently, route rankings should be interpreted alongside route volume and variability.

7. Key Findings

1. Data quality is the most important finding. Order dates range from 2024-01-02 to 2025-12-31, while ship dates range from 2026-06-30 to 2030-06-28.
2. The resulting lead-time range is 904–1,642 days, with an average of approximately 1,320.8 days and a median of 1,274 days.
3. Standard Class has the highest shipment volume at 6,120 records.
4. Gulf has the lowest observed regional average lead time at approximately 1,311.4 days.
5. All 15 products are successfully mapped to one of five factories.
6. The route-level view contains 196 factory-to-state combinations.
7. Small-volume routes should not be interpreted without considering sample size.
8. The interactive threshold makes the definition of a delay explicit and adjustable.

8. Dashboard Design and Implementation

The project implements an interactive Streamlit dashboard using Python. Pandas and NumPy support data transformation and aggregation, while Plotly provides interactive charts and geographic visualization.

The dashboard is organized into four analytical modules:

- Route Efficiency Overview — route averages, route rankings, shipment volume, variability, and efficiency score.
- Geographic Shipping Map — state-level lead time, volume, and sales context.
- Ship Mode Comparison — descriptive comparison of shipping modes and delay behavior.
- Route Drill-Down — selected-route KPIs and order-level shipment details.

User controls include order-date range, region, state/province, ship mode, lead-time threshold, and minimum shipment count. This design allows a stakeholder to move from a network-level overview to a specific route and its underlying transactions.

9. Discussion

The results demonstrate the value of route-level aggregation for logistics analysis. Average lead time alone is not sufficient: a route with one shipment and a route with several hundred shipments should not receive equal operational priority. Combining volume, average lead time, and variability produces a more useful basis for investigation.

The date inconsistency limits the operational interpretation of the current results. The observed lead times are much larger than expected for ordinary parcel shipping. Possible explanations include incorrect year values, date-format interpretation, synthetic/test-data transformations, or historical system offsets. The correct response is to validate the source rather than silently altering the records.

The project also illustrates why descriptive shipping-mode comparisons should not be treated as causal. Differences may reflect geography, product mix, customer type, or order composition. A future study should control for these factors before recommending changes to shipping policy.

10. Recommendations

- Immediate recommendations:
- Validate the Order Date and Ship Date fields with the source system.
- Add automated checks for negative and unusually long lead times.
- Review high-volume routes with poor observed performance first.
- Establish service-level targets by shipping mode.
- Short-term recommendations:
- Add factory capacity, dispatch backlog, carrier, and inventory availability.
- Compare route performance with geographic distance.
- Monitor weekly and monthly trends.
- Develop a composite operational-priority score combining normalized volume and lead time.
- Long-term recommendations:
- Predict expected lead time and late-delivery probability.
- Incorporate carrier, holidays, weather, traffic, inventory, and warehouse capacity.
- Apply optimization techniques for factory allocation and route planning.

11. Future Scope

The current project can be extended into supervised machine learning. Regression models such as Linear/Ridge Regression, Random Forest Regressor, and Gradient Boosting can predict lead time. Classification models such as Logistic Regression, Random Forest, and Gradient Boosting can estimate the probability of late delivery.

Potential features include factory, customer state, region, ship mode, division, units, sales, order month, distance, historical route average, and historical route variability. Regression can be evaluated using MAE and RMSE, while classification can use Precision, Recall, F1-score, and ROC-AUC.

A further extension is route optimization, where predicted service performance and operational constraints are combined to recommend factory assignments or shipping strategies.

12. Conclusion

This study presents a complete data-science workflow for factory-to-customer shipping route analysis at Nassau Candy Distributor. The solution transforms raw transactional data into route-level KPIs, geographic summaries, shipping-mode comparisons, and an interactive Streamlit dashboard. It demonstrates data cleaning, feature engineering, aggregation, visualization, dashboard development, and business interpretation.

The most important conclusion is that data validation must precede operational decision-making. The unusual 2024–2025 order dates and 2026–2030 ship dates generate lead times that are not representative of normal parcel delivery. Once this issue is resolved, the dashboard architecture can be reused for trustworthy logistics monitoring and extended into predictive and optimization-based supply-chain analytics.

13. Limitations

The study is limited by the fields available in the supplied dataset. It does not contain reliable carrier information, factory capacity, actual transportation distance, traffic, weather, inventory availability, warehouse backlog, or confirmed customer delivery dates. Consequently, the analysis identifies associations and descriptive patterns rather than causal explanations.

The date inconsistency is a particularly important limitation. All lead-time-based findings should be treated as provisional until the source dates are confirmed.

14. Selected Descriptive Results

Table 1. Shipment volume by shipping mode.

Ship Mode	Shipments
Standard Class	6,120
Second Class	1,979
First Class	1,548
Same Day	547

Table 2. Regional shipment volume and average lead time.

Region	Shipments	Avg Lead Time (days)
Gulf	1620	1311.4
Pacific	3253	1322.2
Atlantic	2986	1322.7
Interior	2335	1323.1

References

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Note: This document is a research/project paper prepared from the supplied internship files. It is not presented as a peer-reviewed or formally published research article.